Code No: 135AE **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
DATA COMMUNICATION AND COMPUTER NETWORKS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

1.a) Describe in detail about the concept of data transmission and its terminology with necessary example.
b) Explain about ISO-OSI reference model with a neat sketch. [7+8]

2.a) Summarize circuit switching technology implementation in Telephone networks.
b) Write short note on Multiplexing. [8+7]

3.a) Explain why collision is an issue in a random-access protocol but not in controlled access or channelizing protocols?
b) Brief in detail about collision avoidance in Wi-Fi (802.11). [7+8]

4. With the neat sketches, formulate and explain the working principle of CRC with an example. [15]

5. Define and describe Classful Addressing and Classless Addressing. [15]

6.a) How would you demonstrate the challenges in inter-domain routing?
b) With a neat diagram explain distance vector routing protocol. [7+8]

7. How is congestion controlled? Explain in detail about congestion control mechanisms in transport layer. [15]

8.a) Explain Telnet in detail.
b) Illustrate the role of POP3 in electronic mail applications. Explain in detail. [7+8]

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